Derivation

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$$\nabla = \left(\hat{e}_r \frac{\partial}{\partial r} + \frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta} + \hat{e}_z \frac{\partial}{\partial z} \right) \quad ; \quad \underline{v} = \left(v_r \hat{e}_r + v_\theta \hat{e}_\theta + v_z \hat{e}_z \right) \quad ; \quad \underline{\underline{\tau}} = \nabla \underline{v} + (\nabla \underline{v})^T = \begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ & \tau_{\theta\theta} & \tau_{\theta z} \\ & & \tau_{zz} \end{pmatrix}$$

$\nabla \underline{v}$ → This is a dyadic product but w/ del operator

↳ $$(\nabla \underline{v})^T = \begin{pmatrix} \hat{e}_r \frac{\partial}{\partial r}(v_r \hat{e}_r) & \hat{e}_r \frac{\partial}{\partial r}(v_\theta \hat{e}_\theta) & \hat{e}_r \frac{\partial}{\partial r}(v_z \hat{e}_z) \\ \frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta}(v_r \hat{e}_r) & \frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta}(v_\theta \hat{e}_\theta) & \frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta}(v_z \hat{e}_z) \\ \hat{e}_z \frac{\partial}{\partial z}(v_r \hat{e}_r) & \hat{e}_z \frac{\partial}{\partial z}(v_\theta \hat{e}_\theta) & \hat{e}_z \frac{\partial}{\partial z}(v_z \hat{e}_z) \end{pmatrix}$$

(1) $$\hat{e}_r \left[\frac{\partial}{\partial r}(v_r \hat{e}_r) \right] = \hat{e}_r \left[\frac{\partial v_r}{\partial r} \hat{e}_r \right] + v_r \cancel{\frac{\partial \hat{e}_r}{\partial r}}^{0} = \frac{\partial v_r}{\partial r} \hat{e}_r \hat{e}_r \quad ; \quad \frac{\partial \hat{e}_r}{\partial r} = \frac{\partial \hat{e}_\theta}{\partial r} = \frac{\partial \hat{e}_z}{\partial r} = 0 \quad \& \quad \frac{\partial \hat{e}_r}{\partial \theta} = \hat{e}_\theta \; ; \; \frac{\partial \hat{e}_\theta}{\partial \theta} = -\hat{e}_r$$

Similarly,

(2) $$\frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta}(v_\theta \hat{e}_\theta) = \frac{\hat{e}_\theta}{r} \left[\left(\hat{e}_\theta \frac{\partial v_\theta}{\partial \theta} \right) + v_\theta \cancel{\frac{\partial \hat{e}_\theta}{\partial \theta}}^{-\hat{e}_r} \right] = \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} \hat{e}_\theta \hat{e}_\theta - \frac{1}{r} v_\theta \hat{e}_\theta \hat{e}_r$$

$$\hat{e}_r \frac{\partial}{\partial r}(v_\theta \hat{e}_\theta) = \hat{e}_r \left[\hat{e}_\theta \frac{\partial v_\theta}{\partial r} + v_\theta \cancel{\frac{\partial \hat{e}_\theta}{\partial r}}^{0} \right] = \frac{\partial v_\theta}{\partial r} \hat{e}_\theta \hat{e}_r$$

(3) $$\frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta}(v_r \hat{e}_r) = \frac{\hat{e}_\theta}{r} \left[\frac{\partial v_r}{\partial \theta} \hat{e}_r + v_r \cancel{\frac{\partial \hat{e}_r}{\partial \theta}}^{\hat{e}_\theta} \right] = \frac{1}{r} \frac{\partial v_r}{\partial \theta} \hat{e}_r \hat{e}_\theta + \frac{v_r}{r} \hat{e}_\theta \hat{e}_\theta$$

$$(\nabla \underline{v})^T = \begin{pmatrix} \frac{\partial v_r}{\partial r} & \frac{\partial v_\theta}{\partial r} & \frac{\partial v_z}{\partial r} \\ \frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} & \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} \\ \frac{\partial v_r}{\partial z} & \frac{\partial v_\theta}{\partial z} & \frac{\partial v_z}{\partial z} \end{pmatrix}$$

$$\underline{\underline{\tau}} = \nabla \underline{v} + (\nabla \underline{v})^T = \begin{pmatrix} \frac{\partial v_r}{\partial r} & \frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} & \frac{\partial v_r}{\partial z} \\ \frac{\partial v_\theta}{\partial r} & \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} & \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} & \frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z} + \begin{pmatrix} \frac{\partial v_r}{\partial r} & \frac{\partial v_\theta}{\partial r} & \frac{\partial v_z}{\partial r} \\ \frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} & \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} \\ \frac{\partial v_r}{\partial z} & \frac{\partial v_\theta}{\partial z} & \frac{\partial v_z}{\partial z} \end{pmatrix}$$

$$\underline{\underline{\tau}} = \begin{pmatrix} 2\frac{\partial v_r}{\partial r} & \frac{\partial v_\theta}{\partial r} + \frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ & & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$